

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C/C++, Java, SQL

Backend & Systems: FastAPI, Flask, Node.js/Express, React, REST/JSON, WebSockets, real-time streaming, schema design, Docker

Hardware & Manufacturing: CAN/DBC protocols, embedded firmware, VCU and STM32 targets, PX4 flight systems, hardware bring-up, test and validation campaigns

Delivery: pytest, Jest, GitHub Actions, structured logging, software-in-the-loop regression testing, Git/GitHub

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning and technical direction across five project teams spanning controls, telemetry, diagnostics, simulation, and vehicle testing
- Standardize requirements, test evidence, and review process so changes reach the car faster and with fewer regressions

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated flight-control software in C on PX4, using structured telemetry logs to reproduce issues and evaluate changes
- Integrated visual-inertial odometry and iterated configuration through repeatable tests and documented results

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Implemented a real-time control routine in C on an embedded vehicle controller, meeting hard deterministic timing budgets with fixed-point arithmetic
- Built Python telemetry parsers and a software-in-the-loop replay harness that validated changes against recorded logs before they reached hardware

Software Development Intern

Sep 2024 - Oct 2024

Zip Intelligence | Tracy, CA

- Took ownership of a feature from ticket through verified production release, shipping via pull requests with setup and change documentation
- Hardened REST endpoints for status codes and edge cases, and added SQL Server scripts and structured logs to speed reproduction and triage

PROJECT EXPERIENCE

SR-Wireless-CAN — Wireless Flashing & Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, FastAPI, WebSockets, React, SQLite, Docker Compose

- Identified that firmware updates required a tethered vendor tool, decoded the undocumented flashing sequence from bus trace captures, and shipped a Python implementation that flashes the controller over the wireless link
- Led backend development for the platform, building a FastAPI and WebSocket service that streams live telemetry to a browser dashboard with recorded-trace replay
- Implemented structured signal-definition merge with conflict resolution, and containerized the stack with Docker Compose for one-command deployment

Tickr — AI-Coached Paper-Trading Game

2025

TypeScript/JavaScript, React, LLM integration, responsive UI

- Built registration, sign-in, game, and history flows for a scenario-driven trading game with an LLM-backed coach
- Implemented quest and prompt logic, client-side state management, and a mobile-responsive interface